

Article

Modern SCADA for CSP Systems Based on OPC UA, Wi-Fi Mesh Networks, and Open-Source Software

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Abstract: This study presents a methodology for the development of modern Supervisory Control and Data Acquisition (SCADA) systems aimed at improving the operation and management of concentrated solar power (CSP) plants, leveraging the tools provided by industrial digitization. This approach is exemplified by its application to the CESA-I central tower heliostat field at the Plataforma Solar de Almería (PSA), one of the oldest CSP facilities in the world. The goal was to upgrade the control and monitoring capabilities of the heliostat field by integrating modern technologies such as OPC (Open Platform Communications) Unified Architecture (UA), a Wi-Fi mesh communication network, and a custom Python-based gateway for interfacing with legacy MODBUS systems. Performance tests demonstrated stable, scalable communication, efficient real-time control, and seamless integration of new developments (smart heliostat) into the existing infrastructure. The SCADA system also introduced a user-friendly Python-based interface developed with PySide6, significantly enhancing operational efficiency and reducing task complexity for system operators. The results show that this low-cost methodology based on open-source software provides a flexible and robust SCADA architecture, suitable for future CSP applications, with potential for further optimization through the incorporation of artificial intelligence (AI) and machine learning.

Keywords: SCADA; heliostat; OPC UA; Wi-Fi mesh network



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1. Introduction

1.1. Concentrated Solar Power

Concentrated solar technologies (CSTs) are renewable energy technologies focused on concentrating sunlight onto small areas using mainly lenses or mirrors. Inside CST, concentrated solar power (CSP) concentrates solar energy and transforms it into thermal energy at high temperatures. Then, the thermal energy is used to produce steam that drives a turbine connected to an electrical power generator, unlike, for example, photovoltaic (PV) technology, which converts sunlight directly into electricity. In the case of central tower systems, which are the most promising CSP technology, the heliostat is responsible for reflecting and concentrating solar energy on the receiver. Due to the constant Sun movement, the heliostat has to modify its position to reflect and concentrate the solar energy into the receiver and transform it into thermal energy. Central tower systems are composed of several miles of heliostat that have to be controlled individually.

CSP plays a crucial role in the global energy transition for several reasons. First, CSP plants can generate large amounts of clean electricity, making them suitable for utility-scale power generation. The concentrated heat can reach temperatures high enough to produce steam for traditional steam turbine generators, which are a proven and efficient method of generating electricity. Secondly, one of the significant advantages of CSP over other

renewable energy technologies is its ability to store a large amount of thermal energy at a low cost. This storage capability allows CSP plants to generate electricity even when the sun is not shining, such as at night or during cloudy periods. This feature provides a stable and reliable power supply. The ability to store and dispatch power makes CSP an excellent complement to other intermittent renewable energy sources like PV or wind power, helping to stabilize the grid and ensure a continuous power supply.

Despite its advantages, concentrated solar power (CSP) faces significant challenges compared to other major renewable energy technologies. CSP plants generally require higher initial capital investment due to the complexity and scale of the technology, which in turn leads to a relatively high levelized cost of electricity (LCOE) [1]. Several factors contribute to this issue. First, the technological maturity of CSP lags behind that of photovoltaics (PV) and wind power technologies, which have benefited from rapid advancements, economies of scale, and technological innovation. As a result, CSP has not seen the same level of widespread deployment and investment, limiting potential cost reductions through learning curves and mass production. Furthermore, the operation and maintenance (O&M) of CSP plants are more complex than that of PV systems, requiring specialized skills and infrastructure, thus increasing operational costs. This challenge is exacerbated by the relatively limited innovation and experience within the CSP sector. Additionally, CSP technology has been slower to adopt digitalization and automation compared to PV and wind power, missing opportunities for significant reductions in costs and improvements in operational efficiency. The integration of advanced digital tools and automated systems could substantially lower operational and maintenance costs, enhancing overall efficiency and reducing LCOE, as seen in other sectors, for example, power grid management [2]. For example, efficient data processing and the extraction of hidden patterns can dramatically improve knowledge discovery, offering valuable solutions for optimizing system operations. By combining data mining techniques with knowledge discovery, CSP can transition from traditional SCADA systems to intelligent, data-driven frameworks capable of optimal, adaptive, real-time decision-making [3]. Also, modern SCADA systems, leveraging data-driven approaches, can significantly reduce damages through effective alarm management. These systems facilitate the identification of root causes for major failure modes, helping pinpoint critical subsystems. By doing so, they not only optimize maintenance costs but also contribute to design improvements, enhancing their reliability and efficiency over time [4].

1.2. Plataforma Solar de Almería

The Plataforma Solar de Almería (PSA) is a leading global research center focused on solar technology, particularly concentrated solar power (CSP). Located in Spain's Tabernas Desert, PSA benefits from high solar radiation, making it ideal for solar research and innovation. The center plays a key role in CSP development, testing solar thermal components under real-world conditions, and advancing sustainable solar power systems. PSA's research sets industry benchmarks and influences energy policy decisions, with collaborations involving universities, research institutions, and industry partners worldwide [5].

PSA hosts various CSP technologies, including parabolic troughs, solar towers, and Fresnel reflectors. The center is pioneering the integration of artificial intelligence (AI) and the Internet of Things (IoT) to optimize solar power plant operations. A major focus is the CESA-I central receiver system, see Figure 1, which uses 300 heliostats with a surface area of 39.6 m² each. Despite over 20 years of operation, the heliostat field remains in optimal condition due to ongoing maintenance. The system has recently incorporated an AI-based solar tracking system (HelloT), allowing it to operate alongside traditional tracking systems.

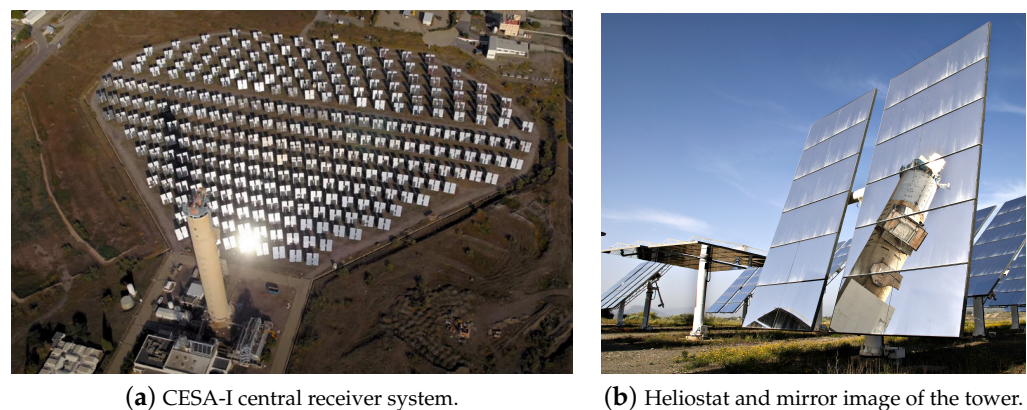


Figure 1. CESA-I system captions

Also for this work, a new Wi-Fi mesh communication system has been successfully deployed for the first time for use in a CSP installation with excellent results; firstly, it allows removing costs in civil works and material due to the elimination of wiring, and secondly, it allows great flexibility in terms of communications. This communication system is used by the newly developed smart heliostat (HelloT).

1.3. Smart Heliostat

The HelloT smart heliostat is an innovative technology designed to optimize solar energy collection through precise and intelligent tracking of the Sun. This system employs Internet of Things (IoT) technology, low-cost hardware, computer vision, and machine learning algorithms to reduce the need for manual interventions and costs, and improve efficiency, reliability, and automation of solar power plants [6]. Furthermore, its decentralized control architecture enables scalability, making it suitable for both small and large-scale solar fields. HelloT also provides highly valuable information to the main control system, further enhancing the performance of the entire plant and reducing costs. This includes predictions on cloud movements and solar energy transients, which enable more efficient management and operation of the solar power plant [7].

The PSA is the main actor developing and testing the HelloT smart heliostat [6]. The research of PSA in this field focuses on the design and implementation of wireless communication technologies, sensor networks, data acquisition platforms, and control algorithms based on neural networks to create a robust and flexible heliostat field capable of autonomous operation under varying environmental conditions. Performance evaluation tests have shown significant improvements in tracking precision and system reliability for the HelloT system [6].

1.4. Supervisory Control and Data Acquisition System in Energy Sector

Supervisory Control and Data Acquisition (SCADA) systems are a key part of plants in the energy sector [2], providing essential monitoring and control processes that are crucial for efficient energy management [8]. SCADA systems—widely used in various energy applications, including power generation, transmission, and distribution [9]—enable real-time data acquisition, system supervision, and automated control. These capabilities are vital for maintaining the reliability of power grids, as they allow for continuous monitoring and fast identification and correction of issues. SCADA systems also support data collection and analysis, which can be used for predictive maintenance and refinement of operational strategies, ultimately enhancing system performance and reducing downtime. However, increasing connectivity and integration of SCADA systems expose them to cyber attacks, which requires the implementation of robust security measures to protect critical infrastructure [10–12].

The recent appearance of open-source SCADA systems and IoT technologies has democratized access to advanced monitoring and control systems, for example, open-source

platforms combined with low-cost hardware and microcontrollers enable the development of low-cost SCADA systems [13]. These systems leverage IoT to provide enhanced connectivity and data acquisition capabilities. These types of solutions not only represent a cost reduction because they are open source, but they also present a continuous updating and evolution by large communities of developers that allow SCADA systems developed with these tools to be kept up to date.

Specifically, CSP plants require the coordination of multiple processes to achieve efficient operation due to the complexity of the system. To manage this complexity, SCADA systems play an essential role in ensuring that each process is synchronized and operated at optimal performance levels. As commented before and despite their relevance, CSP plants have been relatively slow to adopt advanced digitalization tools, and SCADA systems, as a key component of CSP infrastructure, can benefit particularly from digital advancements. Enhanced digitalization can significantly improve the functionality and effectiveness of SCADA systems, offering better integration, more precise control, and improved decision-making capabilities. Embracing these digital tools could greatly enhance the efficiency and reliability of CSP operations. For example, machine learning algorithms can forecast equipment failures in solar power plants, allowing timely maintenance and minimizing downtime. This predictive capability is essential for maintaining high operational efficiency and reducing maintenance costs. In addition, machine learning can offer valuable insight into key operational aspects, such as transients in solar energy. By analyzing these transient variations, machine learning algorithms can improve overall plant performance, mitigate critical situations, extend plant operational lifespan, and further reduce operational costs [7].

In conclusion, CSP has the potential to significantly increase the penetration of renewable energy in the global energy mix, particularly due to its ability to manage power production through thermal energy storage. To fully realize this potential, the CSP must focus on reducing costs and enhancing efficiency. Despite its current challenges, CSP has substantial opportunities for improvement, mainly because it has yet to widely adopt advanced tools from energy digitalization, such as artificial intelligence (AI) and the Internet of Things (IoT). SCADA systems are crucial in the control and operation of CSP plants, particularly given the complexity of these systems. Note that SCADA systems could benefit particularly from digital advancements. Research centers like PSA, which is dedicated to being at the forefront of solar technology, should prioritize the study and development of SCADA systems that integrate the latest digital tools. Incorporating technologies such as HelloIoT into SCADA systems can drive technological advances, improve operational efficiency, and reduce costs. This approach will make CSP a more competitive and reliable source of renewable energy.

For all that, this work presents the development of a new SCADA methodology designed to incorporate cutting-edge digitalization technologies such as AI, IoT, and Wi-Fi mesh networks. Tailored for a research environment, the system is highly flexible, enabling seamless adaptation to new developments such as HelloIoT, experimental setups, new prototypes, storage solutions, and advanced aiming strategies. Using open-source platforms, the system reduces costs. In addition, the proposed SCADA enhances plant monitoring, control, and overall efficiency. This approach significantly contributes to the competitiveness and reliability of CSP technologies in the evolving renewable energy landscape.

The following work is structured as follows: The Materials and Methods section describes the architecture of the proposed SCADA, which includes a server managing information traffic between different components. The Results and Discussion section presents the test and the performance evaluation of the new SCADA implementation for CESA-I. Finally, the Conclusion section summarizes the key findings and emphasizes the potential of the new SCADA system to enhance the competitiveness and reliability of CSP technologies.

2. Materials and Methods

The architecture of the proposed control system is based on a server that is responsible for managing the information traffic between the different actors (see Figure 2).

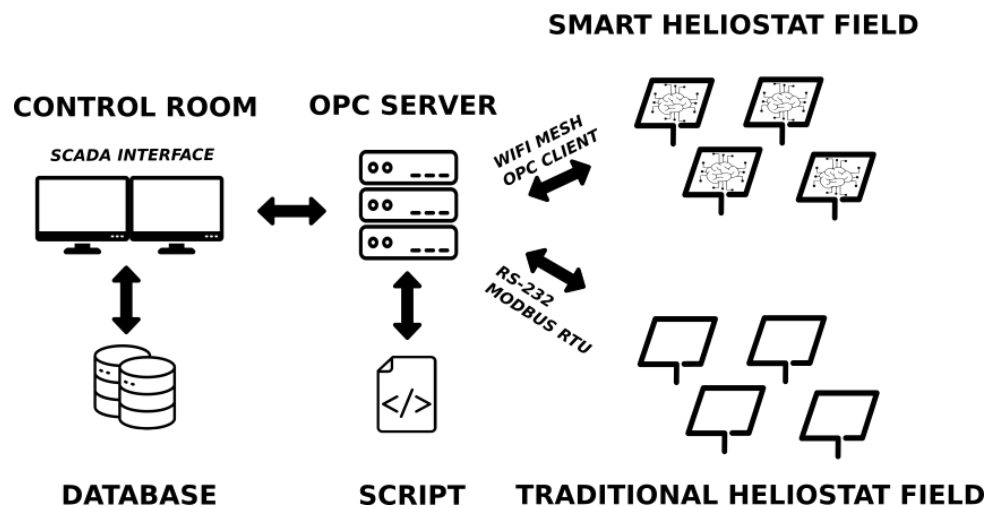


Figure 2. SCADA CESA-I architecture.

In this work, three major developments have been carried out, firstly, the work was conducted to develop the OPC server, which is the key element of the whole project. Secondly, the gateway between the traditional communication system and the server is used to integrate the new developments with the traditional local control. Finally, the SCADA interface allows operators to interact with the CESA-I field.

2.1. OPC Server

As illustrated in Figure 2, the core component of the system is an Open Platform Communications Unified Architecture (OPC UA) server, which acts as a central hub, facilitating communication between the different subsystems.

The OPC UA protocol is a widely adopted client-server communication standard in industrial automation, designed for secure, reliable, and platform-independent data exchange. As the successor to OPC Classic, OPC UA addresses the limitations of platform dependency and enhances interoperability across different systems and networks. Key improvements include platform independence, enhanced security features, and support for complex data types, making OPC UA suitable for both small-scale and large-scale industrial applications [14]. Its service-oriented architecture (SOA) allows modularity and facilitates the integration of additional functionalities. The robust security model incorporates encryption, authentication, and auditing, ensuring data integrity and confidentiality in sensitive industrial environments such as CSP systems.

The present work employs the open-source Python [15] library `opcua-asyncio` [16], an implementation of OPC UA standards that provides asynchronous support for non-blocking operations. Such functionality is crucial for high-performance Industrial IoT (IIoT) systems, where real-time data processing is essential. The server implementation also includes a `SubHandler-Class` that manages subscription-based notifications, enabling efficient handling of events and data changes to provide real-time feedback to the SCADA system. The server leverages Python's `asyncio` library to concurrently manage multiple tasks, a critical feature for maintaining real-time performance in industrial environments. This implementation represents a robust approach to managing a large-scale heliostat field using OPC UA technology.

Using this library, the OPC UA server constructs a hierarchical structure of nodes, each representing one of the 300 heliostats in the CESA-I field. Each device node is an instance of a heliostat object type, encapsulating attributes and behaviors specific to both

the traditional and the smart heliostat. Table 1 shows the structure of the heliostat node (nested variables and their description).

Table 1. Heliostat object type.

Node/Variable	Description
Command_node	A child node of the heliostat node used for sending commands to the heliostat's local control system.
Command	Writable variable that holds the current command to be executed (e.g., sun tracking, stop, cleaning).
Command_op	Writable variable containing additional operational parameters for the command.
Diag	Diagnostic variable covering aspects of the system's diagnostics, such as failure states and local control conditions.
Event	Logs event information related to the heliostat (e.g., setpoint achievements, reference point detections).
HEL_IoT_id	Represents the unique identifier of the heliostat device.
Smart_node	A child node of the heliostat node for advanced operations within the heliostat system.
State	Writable variable storing the current state of the smart node; used to trigger smart operations.
HelloT	Writable variable storing complex data related to the heliostat operations, formatted as a JSON-like string.
State	Holds the operational state information of the heliostat (e.g., sun tracking, stop).
pos_az and pos_el	Store the actual azimuth and elevation positions of the heliostat.
setpoint_az and setpoint_el	Indicate the setpoint positions for the azimuth and elevation axes.

As depicted in Figure 2, the OPC server interacts with various OPC clients. Firstly, the OPC server communicates with the SCADA interface through the PSA communications network, allowing operators to provide or receive information. Additionally, external operators can automate external processes through an application or script through another OPC client. The OPC server also exchanges information with the smart heliostats deployed at CESA-I, which are equipped with OPC clients and communicate over a Wi-Fi mesh network. Finally, the OPC server is connected to traditional heliostats through an OPC-Modbus gateway developed for this work. All data exchanges are handled by the server's event subscription system. This modular and flexible design facilitates the seamless integration of new technologies, such as advanced data analytics or machine learning algorithms, making it highly adaptable to future innovations at PSA.

Furthermore, the architecture incorporates log storage systems, which are essential for maintaining historical data, tracking operator actions, helping in system analysis, and optimizing performance over time. The tracking of server errors is implemented through logging mechanisms, ensuring efficient troubleshooting and system reliability.

2.2. Gateway Modbus-OPC

As commented before, with the main objective of maintaining the existing Modbus-based wired communication and local heliostat controls of the CESA-I field, a gateway has been developed to interface the traditional architecture with the new OPC server. This gateway functions as an application with an integrated OPC client that periodically (less than 2 s) queries the values of the node variables corresponding to each heliostat's local control. On the one hand, if modifications are detected in the local control data, the gateway updates the variable in the corresponding heliostat node in the OPC server by writing the

new values. On the other hand, if a Command child node on the server is altered by another client, the gateway is responsible for transmitting this command to the corresponding local heliostat control via Modbus.

Given the design of the communication network in the traditional heliostat field, where the 300 heliostats are distributed across 16 independent RS-232 communication lines, this communication process consists of 16 parallel threads. Each thread operates continuously, sequentially querying (polling function, see Figure 3) each heliostat on its assigned communication line about the variables defined in the heliostat object type. If a change in any variable is detected, its updated value is sent to the OPC server through the OPC client.

```
def polling(self, reg_name_list = ['ESTADO', 'EVENTOS', 'POSICION_AZ',
    'POSICION_EL', 'CONSIGNA_AZ', 'CONSIGNA_EL', 'DISTANCIA_CENITAL_SOLAR',
    'AZIMUT_SOLAR', 'CORRECCION_REFRACCION', 'HORA_CALCULO_SOLAR'], print_poll = True):
    """ Pregunta al heliostato informacion sobre los registros listados (reg_name_list) """
    polling = {}
    for name in reg_name_list:
        polling.update({name:0})
    for reg in reg_name_list:
        #Posicion y consigna elevacion y acimut
        if self.READABLE_REG[reg] in [20,21,22,23]:
            #value = self.heliostat_modbus.read_register(self.READABLE_REG[reg], number_of_decimals=0, functioncode=3, signed=True)
            value = self.heliostat_modbus.read_register(self.READABLE_REG[reg], signed= True, functioncode=3)
            polling[reg] = value
        #Distancia cenital, azimut solar, correccion refraccion, hora calculo solar
        elif self.READABLE_REG[reg] in [24,26,28,30]:
            value = self.heliostat_modbus.read_float(self.READABLE_REG[reg])
            polling[reg] = value
        #diagnostico az, diagnostico ele
        elif self.READABLE_REG[reg] in [18,19]:
            value = self.diagn(self.READABLE_REG[reg])
            polling[reg] = value
        #estado
        elif self.READABLE_REG[reg] == 16:
            value = self.state()
            polling[reg] = value
        #eventos
        elif self.READABLE_REG[reg] == 17:
            value = self.event()
            polling[reg] = value
        elif self.READABLE_REG[reg] == 9999:
            value = self.heliostat_modbus.read_registers(16,4)
            polling[reg] = value
        else:
            print('Error: Registro no disponible')

    if print_poll:
        print('Polling:')
        print('=====')
        for reg in reg_name_list:
            print(str(reg)+'--> '+str(polling[reg]))
        print('=====')

    return polling
```

Figure 3. CESA-Modbus library polling function.

As mentioned earlier, these 16 processes run cyclically and indefinitely, unless the OPC client is notified of a modification in a server node via a subscription event, such as a change in the command node. In such a case, the corresponding thread stops its routine querying process and sends the updated command to the local control of the specified heliostat. Once the command is transmitted, the querying process resumes.

This loop process is fully implemented in Python and has necessitated the development of a custom Python library, called CESA-Modbus, for communication with the local heliostat controls. This library helps us to abstract from the lower-level programming needed for Modbus communication and is mainly based on the Python library MinimalModbus. Note that, as commented on earlier, the gateway also is equipped with an OPC client, developed using the opcua-asyncio library, which handles events and data exchanges with the OPC server.

2.3. SCADA Interface

A new interface for the SCADA system was developed in this work. Like the previous developments, this was entirely implemented in Python, primarily utilizing the open-source library PySide6. The initial phase of the development involved gathering all the

requirements from the operators of the CESA-I system, some of whom have over 30 years of experience operating these systems. Afterward, a visual style was defined, and all graphical components, windows, and user options were designed. This process was iterated several times in collaboration with the operators, incorporating their suggestions to optimize usability.

Subsequently, all the necessary code to provide functionality to the components was developed. The interface was created as a standalone Python application capable of exchanging information with the OPC server through an integrated OPC client and running on any device within the PSA network, ensuring flexibility and accessibility across the entire infrastructure.

The main window is divided into four areas (see Figure 4): heliostat field, information, control, logging, and console.

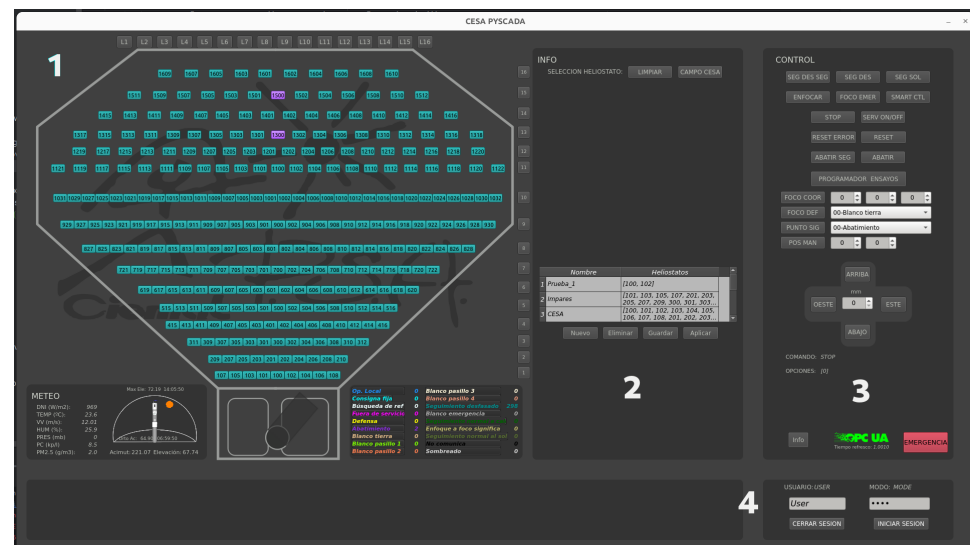


Figure 4. SCADA main window (1—heliostat field, 2—information, 3—control, 4—logging and console).

The primary area, the heliostat field, contains three different sub-areas: meteorological information, a legend, and a central area that represents the CESA-I field.

The meteorological information sub-area (see Figure 5) provides the operator with all the necessary real-time meteorological data, primarily related to solar conditions, required for system operation. This information is obtained from another OPC server at PSA that publishes meteorological data from the extensive network of sensors deployed across PSA. This server was recently introduced as part of PSA's modernization efforts, improving accessibility to the data.

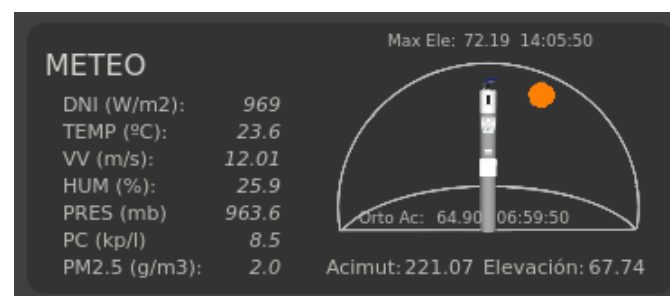


Figure 5. Meteorological data subsection.

As shown in Figure 5, variables such as direct normal irradiation (DNI), temperature, wind speed, humidity, atmospheric pressure, total particles, and particulate matter under 2.5 microns are continuously sampled and updated. Additionally, a dynamic graph provides information on the relative position of the sun (azimuth and elevation), along with the times and values for sunrise and sunset.

The central part of the main area contains a scaled representation of the CESA-I field, where each button corresponds to a heliostat. The button background color indicates the current state of the heliostat, while the color of the text on the button signifies whether the state has been reached (black) or is in transition (green) (see Figure 6).

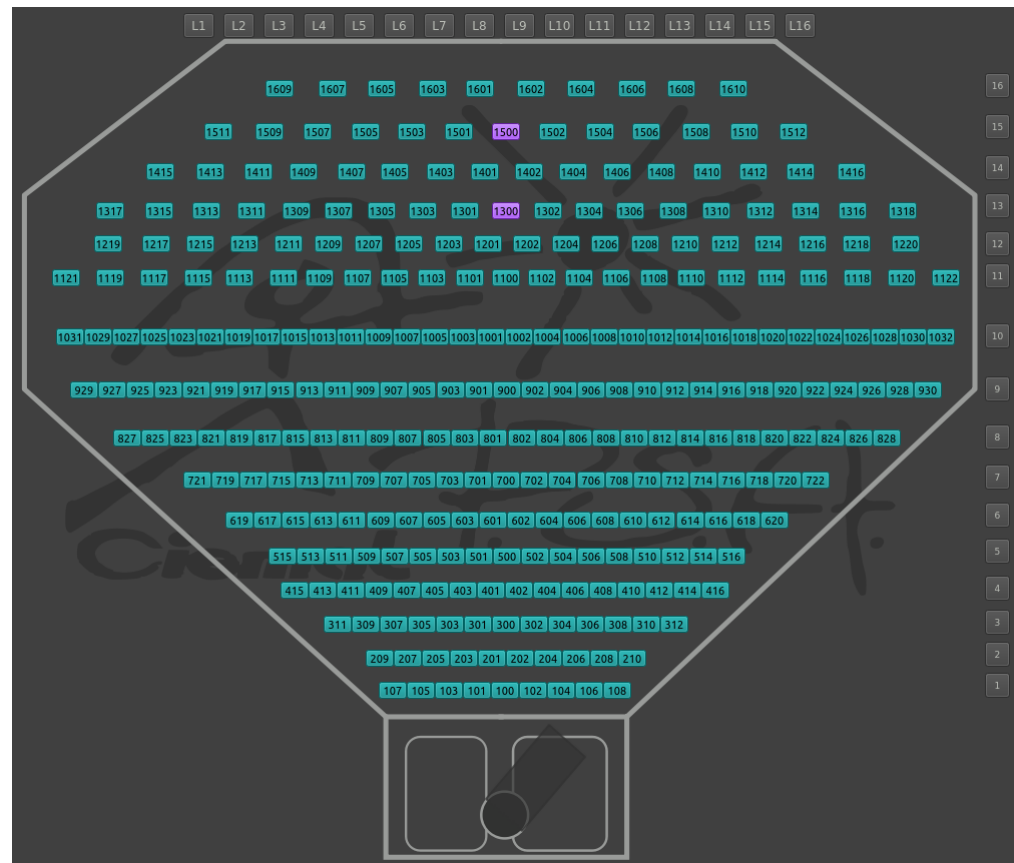


Figure 6. Central subsection of the main area.

In this diagram, individual heliostats can be selected by clicking on the corresponding button, dragging and dropping, or using the top or side rows of buttons to select heliostats by communication lines (L1 ... L16) or rows (1 ... 15), respectively.

At the bottom of the diagram, which represents the tower and auxiliary elements, a rectangular polygon shows the shadow projection over the field. The transparency of the polygon's color indicates the shadow's intensity based on DNI, providing operators with valuable information when selecting heliostats for testing.

A dedicated sub-area within the main interface displays the color legend for heliostat states in Spanish (see Figure 7). Each state is represented by a unique background color for the corresponding buttons, which matches the text color in the legend. Additionally, a counter next to each state indicates the number of heliostats currently in that specific state. Operators can easily select all heliostats in a given state by clicking on the respective text in the legend.

The meaning of each state is as follows:

- Operación local: Heliostat operating under local control only.
- Consigna fija: Heliostat positioned at a fixed setpoint for both azimuth and elevation.

- Búsqueda referencia: Heliostat searching for the ‘zero reference’ in both azimuth and elevation.
- Fuera de servicio: Heliostat that is out of service and not operational.
- Defensa: Heliostat in a defensive position to protect against high winds.
- Abatimiento: Heliostat in its rest or stow position.
- Blanco tierra, pasillo 1, pasillo 2, pasillo 3, pasillo 4: Consecutive safety corridor points used to guide the heliostat toward the receiver.
- Seguimiento desfasado: Heliostat is tracking, but aimed at a standby position near the receiver, not directly at it.
- Blanco emergencia: A designated target point used in emergency situations.
- Seguimiento normal a caldera: Heliostat tracking a specific point at the top of the tower, different from the main receiver.
- Enfoque a foco significativo: Tracking toward a user-defined significant target.
- Seguimiento normal al sol: Heliostat tracking the sun with its aim vector parallel to the solar vector.
- No comunica: Communication with the heliostat has been lost.
- Sombreado: The heliostat is shaded by the tower.

This system ensures that operators can quickly interpret the current state of the heliostats and manage them efficiently.



Figure 7. Legend subsection of the main area.

To the right of the main area is the information section (see Figure 8), which displays the numbers of the selected heliostats and includes a component for managing predefined heliostat groups. This component allows users to create, delete, and load predefined heliostat lists. Additionally, this section features a “CAMPO CESA” (CESA field in Spanish) button that enables the selection of all heliostats in the field at once.



Figure 8. Information area.

The “control” area (Figure 9) is next located, which contains buttons to control commands to either a single heliostat or a group of heliostats, selected using the various methods previously described. Additionally, a text display shows the most recent command and its associated options. At the bottom of the control area, there is an “INFO”

button that opens a window providing information about the interface. An “OPC UA” icon is also visible, indicating the status of communication with the server (green if the connection is active and red in case of failure) along with the refresh time in milliseconds. Finally, there is an “EMERGENCIA” (emergency in Spanish) button, which safely sends all heliostats in the field into stow mode.



Figure 9. Control area.

In the lower right section, the “Login” area (see Figure 10) is dedicated to user management for the interface. In this area, users can log in, log out, and view the currently logged-in user. A user permission control system has been developed to restrict the actions that each user can perform. There are three permission levels: guest, operation, and configuration. Each user is assigned a permission level that limits their allowed actions. The guest level only allows users to consult data, the operator level enables actions related to the routine operation of the field, such as sending heliostats to tracking mode, and the configuration level grants full access to all interface functionalities, including both routine field operations and heliostat configuration settings.

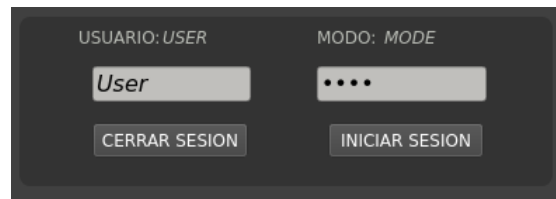


Figure 10. Login area.

Finally, in the lower left section (see Figure 11), the console area displays the most recent actions performed on the interface by the operators. The actions are shown in the following format: date, time, user, permission level, command/action, and options. These logs are also automatically saved to a log file for future reference.

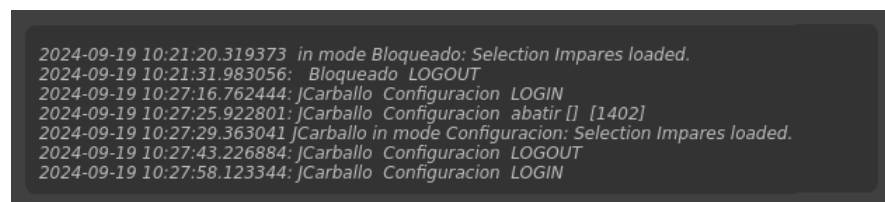


Figure 11. Login area.

Clicking on any button representing a heliostat within the CESA field diagram in the central area opens a new window (see Figure 12), which provides all available information about the selected heliostat, along with operational and control options. These options will be enabled or disabled depending on the permission level assigned to the user currently operating the interface.

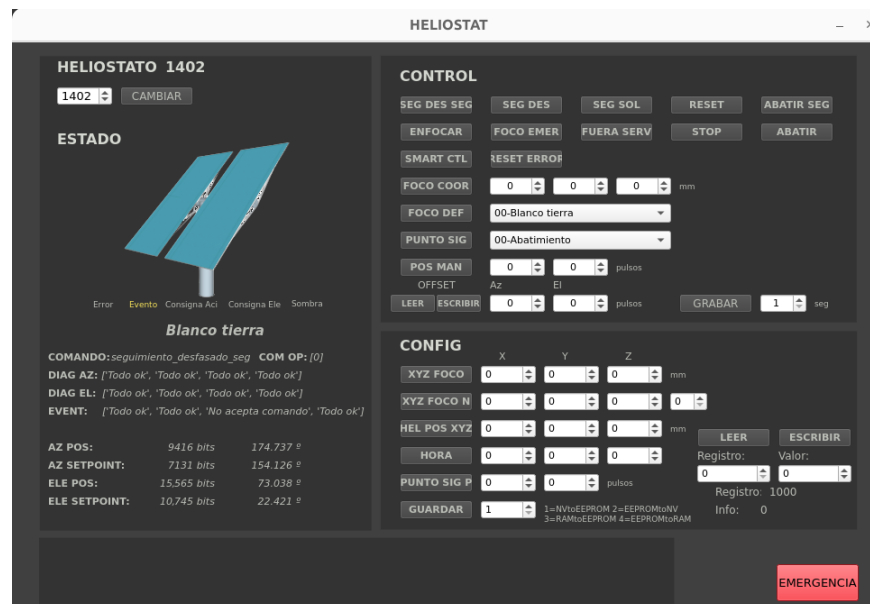


Figure 12. Heliostat window.

The left side of the window is dedicated to displaying relevant heliostat information. Here, an image showing the approximate position of the heliostat is presented, along with five indicator LEDs accompanied by text. These LEDs will remain blank unless a specific condition is met, such as an error, an event, reaching the setpoint in elevation or azimuth, or being in the shadow of the tower, in which case the corresponding text will change its color. Below this section, a larger text field displays the current state of the heliostat, followed by a structured display of all data collected from the local control system during each polling cycle.

The heliostat window contains three additional areas: the control area, which provides all the options related to the normal operation of the heliostat; the configuration area, where parameters of the heliostat's local control can be modified; and a console, similar to the one in the main window, displaying recent actions. Additionally, the window is equipped with an emergency button, identical to the one in the main interface, which sends the heliostat into a safe stow position in case of an emergency.

If, at this point, the “SMART CTRL” button in the control area is clicked, a new window will open (see Figure 13), allowing the initiation of the smart control of the heliostat, provided the heliostat is equipped with the smart control system (HelloT). In this window, in addition to the previously available heliostat information, data related to smart tracking is displayed. This includes images used by the system for tracking, along with neural network detection results, and two graphs showing the tracking error, the setpoint, and the heliostat's position in both azimuth and elevation.

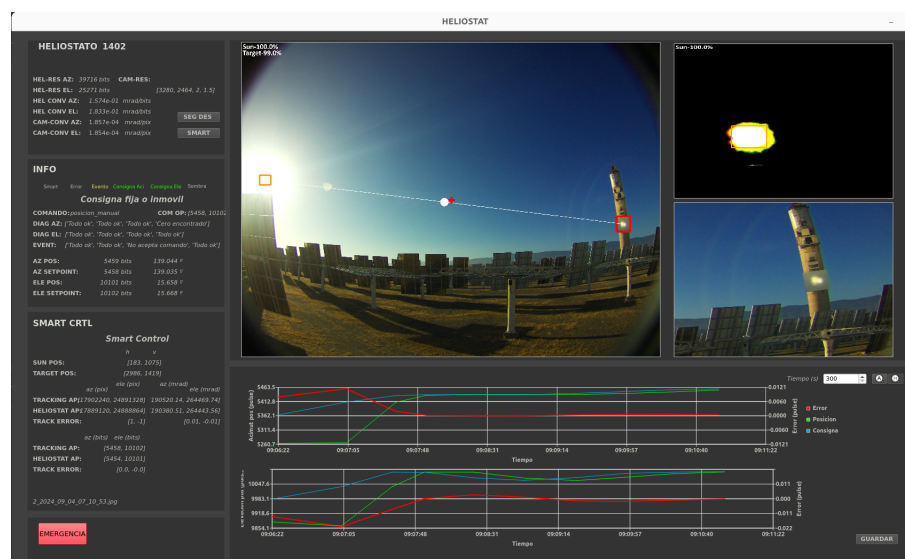


Figure 13. Smart heliostat control window.

3. Results

Various tests, described below, have been conducted on each component of the system to identify and prevent potential operational issues.

3.1. Wi-Fi Mesh Communication Test

The first set of tests focused on the Wi-Fi mesh communication network deployed across the heliostat field. This test aimed to verify the stability and reliability of the wireless communication system under real-time operational conditions. The test involved connecting 10 smart heliostats at different locations within the CESA-I heliostat field and continuously pinging each one during normal operation. The Wi-Fi mesh network successfully maintained stable connections with all smart heliostats distributed across the field, demonstrating reliable communication and network performance under operational conditions. Data transmission rates were consistently high, with minimal packet loss observed (less than 0.1%) even during peak operational periods. Latency remained below 100 ms, ensuring that real-time commands were delivered promptly to the heliostats. The mesh structure also demonstrated robustness against single-point failures, with no significant performance degradation when individual nodes temporarily lost connection. In general, the Wi-Fi mesh system proved to be both resilient and efficient for the intended application. Note that during normal operation, each heliostat is controlled locally and only needs to exchange information if a new setpoint/command is sent or if detailed status information is requested, the latter does not occur frequently during normal operation and is done on an individual and occasional basis.

3.2. Gateway and Modbus-OPC Library Test

The gateway system, responsible for interfacing between the OPC server and the local heliostat control systems via Modbus, was tested for two weeks under full load with all 16 communication lines active before the deployment. The recursive communication process, which polls each of the 300 heliostats sequentially, was run without any problems during the tests. Nowadays, while the system is under normal operation, thanks to the gateway and Modbus-OPC library the state of each heliostat is updated in less than 2 s, well within the acceptable range for the system's real-time requirements. The handling of control commands and status updates between the local controllers and the OPC server occurred without delays or inconsistencies. If inconsistencies or errors are detected, the system handles them correctly without errors that would slow down the process. This demonstrated that Modbus-OPC integration works effectively under full-scale field conditions.

3.3. OPC Server Test

The OPC server, developed using the Python opcua-asyncio library, was tested for scalability, performance, and reliability. Under normal operational conditions, the server efficiently handled the simultaneous connections of the 300 heliostats, operators, and external clients without noticeable performance degradation. Memory usage remained stable, and CPU load was manageable, even when handling high-frequency event subscriptions and real-time data updates, for example, during field start-up or field shut-down. The latency in the server's responses to client requests averaged 100 ms, which is sufficient for real-time SCADA applications. The event subscription system worked as expected, with notifications regarding heliostat status changes being processed immediately and reflected in the SCADA interface. No critical errors were encountered during prolonged test sessions, confirming the reliability and robustness of the implementation of the OPC server.

3.4. Usability

The new SCADA interface was tested for functionality, usability, and stability by operators with over 30 years of experience with previous systems. After several iterations of feedback and refinement, the final version was rated highly in terms of usability, particularly for its intuitive layout and clear depiction of system states. The integration of predefined heliostat groups and the color-coded status legend proved to be especially useful for managing large numbers of heliostats efficiently. Operators reported that the interface simplified routine tasks, such as sending heliostats into tracking mode or managing emergency situations. User login functionality has been well received, by the head of operation, due to the permission-based control system with appropriately blocked restricted actions based on user roles. The system's emergency response mechanism was also tested, and the "Emergency" button successfully triggered the safe stow procedure for all heliostats without delay. In general, the operators expressed satisfaction with the efficiency and responsiveness of the system. The visual clarity of the interface, the feedback in real-time, and the functionality of the interface were well received. During usability testing, the average task completion times were reduced compared to the legacy system, particularly in heliostat group management and status monitoring tasks.

3.5. Stability

The system's overall stability was evaluated during a 72-h continuous test under normal operating conditions before the final deployment without crashes. The interface successfully exchanged information with the OPC server, and all control commands from the interface were accurately transmitted to the heliostats. The graphical representation of the CESA-I field and heliostat statuses is updated in real-time with minimal latency, providing operators with clear and current information.

The results of the series of tests indicate that the developed SCADA system meets the performance, usability, and stability requirements for real-time heliostat field management.

The Wi-Fi mesh communication, gateway, and Modbus-OPC integration proved to be reliable, while the OPC server maintained robust performance under heavy loads. The new SCADA interface improved operational efficiency, with positive feedback from experienced operators, confirming its effectiveness for industrial-scale solar plant control. In addition, the whole system is still under development, open to changes and improvements, either at the suggestion of the operators or because of the need for further development.

Since the tests were completed, the SCADA system was adopted as the main control and supervision system of the CESA-I field by the operators, abandoning the traditional one, and was put into production.

4. Discussion

The results of the tests conducted on the CESA-I SCADA system indicate significant advancements in the operation, control, and monitoring of the heliostat field. These results align with previous studies on the integration of SCADA systems in solar thermal plants but with notable improvements in terms of costs, flexibility, scalability, and real-time performance. The implementation of a Wi-Fi mesh network, combined with the OPC UA server, provides a robust framework for handling the complexities of large-scale heliostat control, which has been a persistent challenge in the CSP sector.

The successful deployment of the Wi-Fi mesh communication system, as highlighted in Section 3.1, confirms its viability to replace traditional wired communications in CSP plants. This innovation reduces infrastructure costs and enhances the flexibility of heliostat deployment and maintenance, a finding consistent with earlier research that demonstrated the advantages of wireless communication in industrial automation. The low latency and minimal packet loss achieved in the tests are particularly encouraging, suggesting that the Wi-Fi mesh can support real-time solar tracking and control operations. This aspect marks a significant improvement over the limitations of wired systems, which have traditionally constrained the adaptability and scalability of heliostat fields.

The gateway system linking Modbus-based local controls with the OPC UA server was extensively tested, with results that demonstrated seamless integration between legacy systems and modern SCADA architecture. This is an important step forward in maintaining the operational continuity of older heliostat fields while benefiting from modern data acquisition and control technologies. The recursive communication process, which efficiently updates heliostat statuses and commands via OPC-UA, showed no signs of delay or failure, validating the effectiveness of the custom CESA-Modbus library developed for this purpose.

The ability of the OPC UA server to manage the simultaneous connections of more than 300 heliostats, external clients, and operator stations without significant performance degradation demonstrates the scalability and robustness of the server. This confirms the viability of OPC UA for CSP plant operations, a key finding that addresses the gaps in previous studies, which often highlighted the difficulties of managing such extensive data streams in real-time (SCADA). The efficient handling of event subscriptions, with latency averaging around 100 ms, further supports the suitability of OPC UA for industrial control systems, ensuring timely updates and operational safety.

The usability tests of the new SCADA interface, conducted with operators who have decades of experience with the CESA-I field, offer strong evidence of its practical efficiency. The positive feedback on the interface's design and real-time response underscores the importance of operator-centered design in industrial SCADA systems. Furthermore, the stability of the system during prolonged testing, with no crashes or failures, confirms its reliability for continuous operation, an essential requirement for solar thermal plants.

The results highlight several areas for future research. First, combining AI and machine learning with SCADA systems presents a promising path for predictive maintenance and operational optimization, as highlighted in previous studies. AI-driven analytics could enable the system to forecast equipment failures and enhance operational efficiency by analyzing heliostat performance and environmental conditions in real-time. Furthermore,

integrating cloud-based SCADA systems with edge computing could revolutionize scalability and responsiveness. Edge computing would allow data processing to occur closer to the heliostats, minimizing latency for real-time control, while the cloud would provide scalable resources for centralized data management and advanced analytics, making the system capable of efficiently managing even larger heliostat fields.

5. Conclusions

SCADA methodology presented represents a significant step forward in the control and management of solar thermal power plants. The results demonstrate the system's scalability, reliability, and cost-effectiveness. Key outcomes include stable communications with low latency, robust real-time control, and an intuitive user interface that streamlines operational tasks. The implementation of the Wi-Fi mesh network significantly reduces infrastructure costs and enhances flexibility in managing the heliostat field. Additionally, the SCADA system is designed as an open and adaptable platform, offering the potential for future optimizations through artificial intelligence and machine learning. Overall, this methodology strengthens the competitiveness and feasibility of CSP technologies within the renewable energy sector.

Further research should focus on the integration of advanced AI algorithms and the potential for global deployment in other CSP facilities. Also, future works include receiver control, storage management, and smart aiming point strategy [17] integration in the SCADA.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	artificial intelligence
CSP	concentrated solar power
CST	concentrated solar technology
DNI	direct normal irradiance
HelloT	smart heliostat
IoT	Internet of Things
IIoT	Industrial Internet of Things
LCOE	levelized cost of energy
LED	light-emitting diode
OPC	open platform communications
PSA	Plataforma Solar de Almería
PV	photovoltaics
SCADA	Supervisory Control and Data Acquisition
UA	Unified Architecture

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